

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 3

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 4 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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PHYS10302 Vibrations and Waves

Week 3 Questions

1. A system has resonant frequency $\omega_0 = 500 \text{ rad/s}$ and quality factor $Q = 20.0$. It is subject to a force of angular frequency ω which can be varied. If the amplitude at $\omega = 500 \text{ rad/s}$ is 10.0 mm , calculate the amplitude at (a) $\omega = 510 \text{ rad/s}$, (b) $\omega = 600 \text{ rad/s}$, (c) $\omega = 490 \text{ rad/s}$.
2. In the lectures we derived the amplitude of forced oscillations in terms of the angular frequency of the driving force,

$$A(\omega) = \frac{F_0/m}{((\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2)^{1/2}},$$

where $\gamma = b/m$. For $A(\omega)$ to reach a maximum value the denominator must have a minimum value. Hence differentiate the denominator to find an expression for the driving angular frequency at which the amplitude peaks in terms of the quality factor $Q = \omega_0/\gamma$.

3. A mass of 2 kg on a table is attached at one end to a spring of spring constant $k = 300 \text{ N/m}$.
 - (a) What are the natural angular frequency and period of the system (working to 3 significant figures)?
 - (b) The mass is subject to an external driving force in the horizontal direction, with functional form $F = F_0 \cos \omega t$, where $F_0 = 1 \text{ N}$ and $\omega = 12.6 \text{ rad/s}$. The damping constant is $b = 2 \text{ kg/s}$. Compute the amplitude of displacement and the phase angle of the system in steady state oscillations.
4. A series RLC circuit has $C = 8.0 \times 10^{-6} \text{ F}$, $L = 2.0 \times 10^{-2} \text{ H}$, $R = 7.5 \Omega$ and is driven by a voltage of $V(t) = 15 \cos \omega t \text{ V}$. What is the resonant frequency in Hz of the circuit (where the impedance is minimised) and what is the amplitude of the current at this frequency?
5. Look up Barton's Pendulum, a famous resonance experiment, on the internet. Find and watch a video of the experiment, for example the one at <http://www.youtube.com/watch?v=kODOL-QBzSM> and pay close attention to the phase relationship between the driving pendulum and the driven pendula.

E&M Tutorial Sheet 3: Electric Potential

- 1) (Revision) Calculate the electric field at a perpendicular distance z from the axis of a uniform ring of charge, q , with radius a . Why can you not (usefully) use Gauss' Law to solve this question?

- 2) Calculate the electric potential corresponding to each of these electric fields:

(a) $\mathbf{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$ Point charge q

(b) $\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{x}{(x^2 + a^2)^{3/2}} \hat{\mathbf{x}}$ On axis ring of charge q radius a (see Q1)

(c) $\mathbf{E} = \frac{2q}{4\pi\epsilon_0 a^2} \left(\frac{x}{\sqrt{x^2}} - \frac{x}{\sqrt{(x^2 + a^2)}} \right) \hat{\mathbf{x}}$ On axis disc of charge q radius a

Hint: Use infinity as the reference point for zero of electric potential. The potential at a point is defined by $\phi = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l}$ but to make life easier you can integrate in the opposite direction and multiply by -1 and instead use $\phi = \int_r^{\infty} \mathbf{E} \cdot d\mathbf{l}$.

- 3) In E&M Tutorial Sheet 2 you found that the electric field due to a uniform charge density sphere of radius a is given by:

$$r \geq a: \mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$$

$$r < a: \mathbf{E} = \frac{Qr}{4\pi\epsilon_0 a^3} \hat{\mathbf{r}}$$

where $Q = \frac{4}{3}\pi a^3 \rho$. Find the electric potential at the following points:

- (a) At a radius, r , outside of the sphere.
 (b) At the surface of the sphere.
 (c) At a radius, r , inside the sphere.

Hint: When you perform the line integral to a point inside the sphere, do it in two steps; first to the surface of the sphere, then from the surface to the point of interest, substituting the correct form of the field for inside and outside.

- 4) Sketch both E and ϕ for each of the fields in Q2 and Q3.

- 5) The dielectric strength of a material is the highest field that can be sustained without electric breakdown; air becomes ionised and conducting for an electric field that is higher than about $3 \times 10^6 \text{ Vm}^{-1}$. Use the results of Q3 to show that for a charged sphere, the potential and field strength at the surface are related by $\phi = Ea$. What is the maximum potential that can be achieved on a spherical conductor radius 5 cm in air? *This is similar in size to a classroom Van de Graaff generator.* If you wanted to hold a potential of 30MV, how big would the sphere have to be? *This is the maximum voltage that has ever been made artificially and was achieved at a very large tandem Van de Graaff accelerator at Oak Ridge in the US, shown on the right.* To achieve 30MV, the sphere is surrounded with an insulating blanket of sulphur hexafluoride gas, which has a dielectric strength about three times that of air at atmospheric pressure. How would this have changed the radius of the sphere?



- 6) Use the dielectric strength of air given in the previous question to estimate the total charge of a thundercloud. Make any assumptions that seem reasonable. *Hint: it is tempting to model the thundercloud as a sphere, but would a different simple shape be a better model?*

PHYS10372 Mathematics 2: Tutorial-03 Questions

These questions cover the material from week-3 of PHYS10372: Laplacians and vector manipulations of div and curl.

1. Prove the following formulae where ψ and ϕ are scalar fields:

$$\begin{aligned}\nabla(\psi\phi) &= \psi\nabla(\phi) + \phi\nabla(\psi) \\ \nabla(f(\psi)) &= f'(\psi)\nabla(\psi).\end{aligned}$$

2.

(a) $\mathbf{r}$ is a position vector from the origin: $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ and $f(r)$ is a scalar function of r (i.e. $|\mathbf{r}|$).

What are the possible directions of the field lines of the vector-field: $f(r)\mathbf{r}$.

Show that:

$$\nabla \times (f(r)\mathbf{r}) = \nabla f(r) \times \mathbf{r}$$

In which direction does $\nabla f(r)$ point? Hence show that $\nabla \times (f(r)\mathbf{r}) = 0$

b) Sketch field lines for the field $g(r)\hat{\mathbf{k}} \times \mathbf{r}$, where $g(r)$ is a scalar function of r .

Show that:

$$\nabla \cdot (g(r)\hat{\mathbf{k}} \times \mathbf{r}) = \mathbf{r} \cdot (\nabla g(r) \times \hat{\mathbf{k}})$$

and hence that:

$$\nabla \cdot (g(r)\hat{\mathbf{k}} \times \mathbf{r}) = 0$$

3. Let $\mathbf{a}$ be a constant vector and $\mathbf{r}$ be a position vector from the origin.

Use the identity $\nabla \times (\phi\mathbf{A}) = \phi(\nabla \times \mathbf{A}) + (\nabla\phi) \times \mathbf{A}$ to show that:

$$\nabla \times (\nabla \times (r^2(\mathbf{a} \times \mathbf{r}))) = \nabla \times (r^2(\nabla \times (\mathbf{a} \times \mathbf{r})) + (\nabla r^2) \times (\mathbf{a} \times \mathbf{r}))$$

Use the identities:

$$\nabla \times \mathbf{r} = \nabla \times \mathbf{a} = 0 \quad \text{and} \quad \nabla \times (\mathbf{a} \times \mathbf{r}) = 2\mathbf{a}$$

which were derived in earlier tutorial/workshop questions, to prove that:

$$\nabla \times (\nabla \times (r^2 (\mathbf{a} \times \mathbf{r}))) = -10(\mathbf{a} \times \mathbf{r}).$$

This shouldn't take more than 5-10 lines and gives practice in using the various vector identities.

You should work out ∇r^2 and $\nabla(\mathbf{a} \cdot \mathbf{r})$ in terms of $\mathbf{r}$ and $\mathbf{a}$ before starting.

4. Calculate the Laplacian of $\log(x^2 + y^2 + z^2)$ in both Cartesian and spherical polar coordinates. The results should be identical.